PROBLEM 29

Heavy Particle on a Tilted Cylinder

Original statement · Problems.pdf, p. 17

Unilateral Liftoff

A heavy particle of mass m is constrained to move without friction along the surface of an infinitely-long, circular cylinder of radius R. The cylinder axis (Oz) forms an angle α with the vertical direction (Oz_1). The position of the particle is completely specified by providing the coordinate z along the axis Oz and the angle θ. Initially, the particle is located at $z = 0, \theta = \pi/2$ and the time derivatives are $\dot{z} = v_0$ and $\dot{\theta} = \omega_0$.

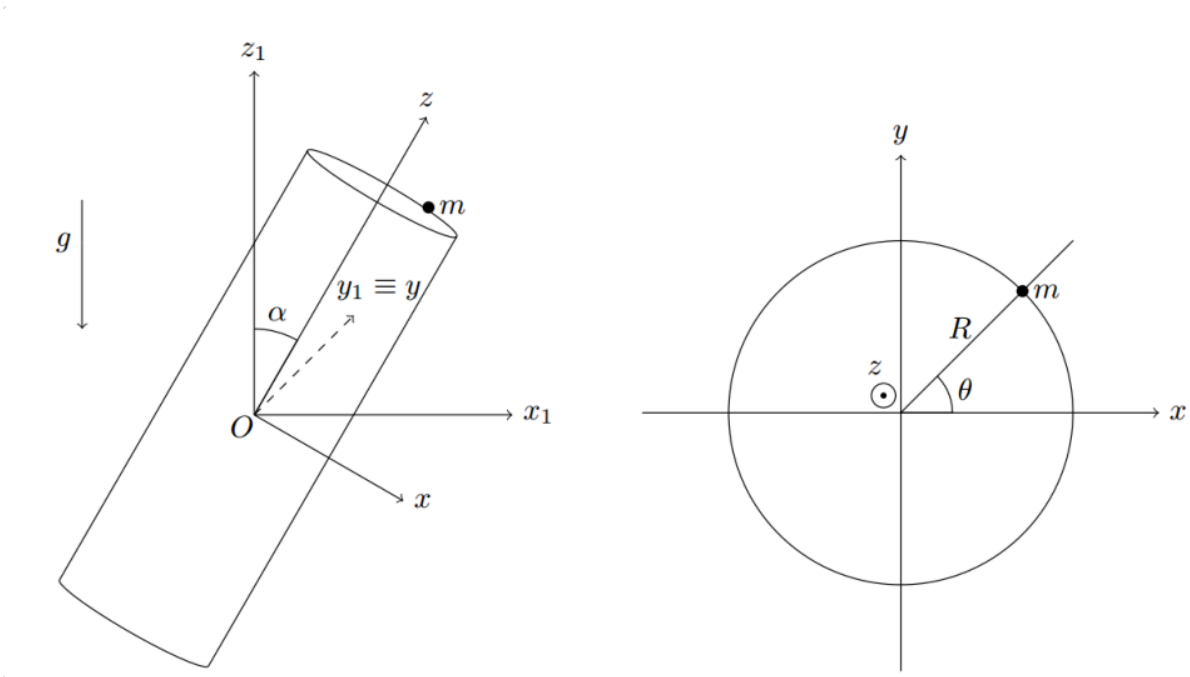


Figure: Left: cylinder with axis Oz tilted by α from the vertical Oz1; right: cross-section with radius R and angle θ (Problems.pdf, p. 17).

(a) Determine the value of z as a function of time, $z(t)$.

(b) Reduce the determination of the angle $\theta(t)$ to integrals of the form $\int [...] d\theta = \int [...] dt$.

(c) Determine the reaction from the cylinder as a function of θ.

(d) For the particular case $\omega_0^2 = \frac{g \sin\alpha}{R}$, describe qualitatively the motion of the particle.

(e) For the particular case $\omega_0^2 = \frac{4g \sin\alpha}{R}$, describe qualitatively the motion of the particle.

Full Analytical Solution (4-Phase Methodology)

PHASE 1

Physical Statement, Hypotheses & Parameters

- **Physical System:** A particle of mass m moves without friction on an infinitely long circular cylinder of radius R whose axis Oz forms an angle α with the vertical.
- **CDOFs:** 2 degrees of freedom: axial coordinate $z(t)$ and azimuthal angle $\theta(t)$.
- **Initial State ($t = 0$):** $z(0) = 0, \theta(0) = \pi/2$, with $\dot{z}(0) = v_0$ and $\dot{\theta}(0) = \omega_0$.
- **Parameters:** $m, R, \alpha, g, v_0, \omega_0$.

PHASE 2

Reference Frames, Vector Bases & Coordinate Geometry

- Cylindrical basis on cylinder: $\mathbf{e}_r, \mathbf{e}_\theta, \mathbf{k}$.
- Gravity vector decomposed along cylinder axes:
- Axial component along $-Oz$: $g_z = -g\cos\alpha$.
- Transverse horizontal/vertical component in cross-section: $g_\perp = g\sin\alpha$.

$$\mathbf{g} = -g\sin\alpha\sin\theta\,\mathbf{e}_\theta + g\sin\alpha\cos\theta\,\mathbf{e}_r - g\cos\alpha\,\mathbf{k}$$

From the figure, $\mathbf{z} = \sin\alpha\,\mathbf{i}_1 + \cos\alpha\,\mathbf{k}_1$ and $\mathbf{x} = \cos\alpha\,\mathbf{i}_1 - \sin\alpha\,\mathbf{k}_1$, so $-g\mathbf{k}_1 = g\sin\alpha\,\mathbf{x} - g\cos\alpha\,\mathbf{z}$: $\theta = 0$ is the lowest generator of the cylinder.

PHASE 3

Step-by-Step Mathematical Deduction

Part (a): Axial Motion $z(t)$

> **Pedagogical Justification:** The cylinder surface is smooth, so the reaction force is purely radial ($\mathbf{N} = N\mathbf{e}_r$) and has zero component along the cylinder axis Oz. The axial dynamics completely decouple into constant gravitational acceleration.

$$m\ddot{z} = -mg\cos\alpha \implies \ddot{z} = -g\cos\alpha$$

Integrate once with $\dot{z}(0) = v_0$:

$$\dot{z}(t) = v_0 - g\cos\alpha\, t$$

Integrate for position with $z(0) = 0$:

$$z(t) = v_0 t - \frac{1}{2} g\cos\alpha\, t^2$$

Part (b): Angular Evolution $\theta(t)$ Reduced to Quadratures

Azimuthal equation:

$$mR\ddot{\theta} = -mg\sin\alpha\sin\theta \implies \ddot{\theta} + \frac{g\sin\alpha}{R}\sin\theta = 0$$

Multiply by $\dot{\theta}$ and integrate:

$$\frac{1}{2}\dot{\theta}^2 - \frac{g\sin\alpha}{R}\cos\theta = C_\theta$$

At $t = 0, \theta = \pi/2 \implies \cos(\pi/2) = 0, \dot{\theta}(0) = \omega_0$:

$$C_\theta = \frac{1}{2}\omega_0^2$$

$$\dot{\theta}^2 = \omega_0^2 + \frac{2g\sin\alpha}{R}\cos\theta$$

Separation of variables:

$$\int_{\pi/2}^{\theta(t)} \frac{d\theta'}{\sqrt{\omega_0^2 + \frac{2g\sin\alpha}{R}\cos\theta'}} = \pm t$$

Part (c): Normal Reaction Force $N(\theta)$

Radial equation along $-\mathbf{e}_r$ (N is the inward reaction; gravity has component $+mg\sin\alpha\cos\theta$ along $+\mathbf{e}_r$, hence $-mg\sin\alpha\cos\theta$ along $-\mathbf{e}_r$):

$$mR\dot{\theta}^2 = N - mg\sin\alpha\cos\theta$$

$$N(\theta) = mR\dot{\theta}^2 + mg\sin\alpha\cos\theta = mR\left(\omega_0^2 + \frac{2g\sin\alpha}{R}\cos\theta\right) + mg\sin\alpha\cos\theta$$

$$N(\theta) = mR\omega_0^2 + 3mg\sin\alpha\cos\theta$$

Part (d): Case $\omega_0^2 = \frac{g\sin\alpha}{R}$

$$\dot{\theta}^2 = \frac{g\sin\alpha}{R}(1 + 2\cos\theta)$$

Turning point occurs when $\dot{\theta} = 0 \implies \cos\theta = -1/2 \implies \theta = \pm\frac{2\pi}{3}$.

The particle undergoes bounded angular oscillations between $-2\pi/3$ and $+2\pi/3$ about the bottom generator $\theta = 0$ (for $\omega_0 > 0$ it first reaches $2\pi/3$, then swings through $\theta = 0$ to $-2\pi/3$ and back) while accelerating parabolically along the axis Oz.

> **Erratum:** The first version had $\mathbf{g}\cdot\mathbf{e}_r = -g\sin\alpha\cos\theta$ and an oscillation between $\pi/3$ and $2\pi/3$. From the figure $\mathbf{g}\cdot\mathbf{e}_r = +g\sin\alpha\cos\theta$ and $\mathbf{g}\cdot\mathbf{e}_\theta = -g\sin\alpha\sin\theta$, so $\theta = 0$ is the stable bottom and the motion in (d) is symmetric about it. The result of (c) was already the inward reaction $N = mR\omega_0^2 + 3mg\sin\alpha\cos\theta$. Numeric check: RK4 of $\ddot{\theta} = -(g\sin\alpha/R)\sin\theta$ from $\theta(0) = \pi/2$ with $\omega_0^2 = g\sin\alpha/R$ gives $\theta_{min}/\pi = -0.6667$ and $\theta_{max}/\pi = +0.6667$; for $\omega_0^2 = 4g\sin\alpha/R$ the particle revolves without stopping.

Part (e): Case $\omega_0^2 = \frac{4g\sin\alpha}{R}$

At the top of the cylinder ($\theta = \pi$, $\cos\pi = -1$):

$$\dot{\theta}^2(\pi) = \frac{4g\sin\alpha}{R} - \frac{2g\sin\alpha}{R} = \frac{2g\sin\alpha}{R} > 0$$

The angular velocity never vanishes. The particle performs complete revolutions (helical trajectory with growing pitch) around the cylinder.

PHASE 4

Physical Interpretation & Dimensional Verification

- Complete decoupling between 1D uniform acceleration along z and nonlinear pendulum dynamics in θ.

Source: Problems.pdf (Page 17)